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A transparent financial risk factor library

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Abstract

Financial risk factor time series are essential for empirical asset pricing research, yet existing factor libraries suffer from limited reproducibility, lack of modularity, and language-specific implementations. This paper presents a comprehensive, fully replicable multilingual financial risk factor library that addresses these limitations. We provide a modular framework for constructing risk factors from raw data, encompassing data acquisition, cleaning, and portfolio aggregation. The library includes both an application programming interface (API) and native implementations in Python and R, allowing researchers to work in their preferred computational environment. Our transparent, modular design enables both reproducibility and flexibility, accommodating diverse portfolio sort configurations and adjustment methodologies. By making all code and data workflows publicly available, we promote standardization and support reproducible research practices in empirical asset pricing.

Keywords: financial risk factors, reproducible research, asset pricing, API, R, Python.

1. Introduction: Constructing financial risk factors

In empirical finance, a key question is which sources of systematic risk drive asset returns. Researchers address this by constructing factor returns—portfolios isolating single economic characteristics or exposures. Factor returns allow model evaluation, hypothesis testing, benchmarking, and decomposition of returns into systematic and idiosyncratic components (Fama and French 1993; Cochrane 2011; Harvey, Liu, and Zhu 2016).

Despite their importance, no widely adopted, transparent, and accessible source standardizes factor construction. Current practice often relies on ad-hoc publication of factor series,

usually without code or detailed workflows. This limits reproducibility, comparability, and extension. Reproducible research requires transparent workflows, standardized data processing, analysis-ready series, and modularity to vary methodological choices.[1]

Our project makes four contributions. First, we release a public factor library that centralizes series and workflows. Second, we provide a modular, end-to-end workflow for data acquisition, cleaning, variable construction, and portfolio aggregation. Third, we offer native R and Python implementations and a programmatic API for standardized data retrieval. Fourth, all code, workflows, and documentation are published to enable replication, methodological comparison, and extensibility. Together, these improve transparency, reproducibility, and comparability.

We provide two libraries — *tidyfinance* (on CRAN) and *py-tidyfinance* that streamline data acquisition from WRDS, Compustat, and other commonly used data providers. Automated routines handle standard lagging procedures to eliminate ad-hoc fixes. Special adjustments typical for financial data —delistings, price or volume adjustments —are embedded.

A core focus of our project is the handling of the construction of portfolio sorts which involve many subjective choices: breakpoints, weighting, rebalancing, handling ties, and sample inclusion. Our code treats sorting as a configurable module. All choices are documented, tested and common variants (equal- or value-weighted, double sorts, quantile/percentile breaks, overlapping portfolios) are implemented.

We illustrate the full workflow in R, from raw data to factor returns, including diagnostics. Python examples in the appendix ensure parity in configuration, defaults, and outputs.

We publish a curated factor library with preprocessed time series across sorting variables, weighting schemes, and adjustments. We demonstrate how methodological choices affect factor returns to highlight empirical relevance. Finally, a programmatic API provides access to standardized series and preprocessed panels. The database supports versioning and batch retrievals. All software, workflows, and documentation are published to enable replication, sensitivity analysis, and extension, promoting standardized benchmarking and transparent empirical research.

2. Vignette: A case study in factor construction

Our focus is on presenting `tidyfinance`. You can install the released version of `tidyfinance` from CRAN via:

```
install.packages("tidyfinance")
```

You can install the development version of `tidyfinance` from GitHub via:

```
# install.packages("pak")
pak::pak("tidy-finance/r-tidyfinance")
```

```
library(dplyr)
```

[1] Existing resources illustrate this gap: the French Data Library (Fama and French 2023) provides canonical factors but no code; the Open Source Asset Pricing Project (Chen and Zimmermann 2022) focuses on replication; the global factor database (Jensen, Kelly, and Pedersen 2023) is Python-oriented. A modular, multilingual framework is still needed.

```
library(tidyr)
library(lubridate)
library(ggplot2)
library(tidyfinance)
```

2.1. Data sources

For our example, we rely on two datasets: CRSP monthly stock returns and Compustat annual financial statement data. We use `tidyfinance` to download and preprocess these datasets, ensuring that they are ready for factor construction. The code snippet below demonstrates how to acquire the necessary data for our analysis.

TODO Describe data sources: CRSP, Compustat, etc and easy access via tidyfinance packages.

```
date_start <- as.Date("1960-01-01")
date_end <- as.Date("2024-12-31")

crsp_monthly <- download_data(
  domain = "wrds",
  dataset = "crsp_monthly",
  version = "v2",
  start_date = date_start,
  end_date = date_end,
  add_ccm_links = TRUE
)

compustat_annual <- download_data(
  domain = "wrds",
  dataset = "compustat_annual",
  start_date = date_start,
  end_date = date_end,
  additional_columns = c("at", "ib"),
  only_us = TRUE
)
```

TODO: Explain all the cleaning steps in the background: remove canadians, etc.

Table 1 provides an overview of all data sources accessible via the `download_data()` interface in both `tidyfinance` and `py-tidyfinance`.

TODO highlight that the same functionality is available in Python via py-tidyfinance. -

Source	Description	Example
Ken French's Data Library	Fama-French factor models, portfolio returns, and industry classifications (169+ datasets at daily/weekly/monthly frequencies)	`download_data("famafrench", "Fama/French 5 Factors (2x3) [Daily]")`

Source	Description	Example
Global Q	Hou-Xue-Zhang Q5 factor model (market, size, investment, ROE, expected growth) at multiple frequencies	`download_data("globalq", "factors_q5_monthly")`
Goyal-Welch Macro Predictors	Macroeconomic predictors: dividend-price ratio, earnings-price ratio, book-to-market, term spread, default yield spread, T-bill rate, inflation	`download_data("macro_predictors", "monthly")`
FRED	Any economic time series from the St. Louis Fed (GDP, CPI, unemployment, etc.)	`download_data("fred", series = c("GDP", "CPIAUCNS"))`
Yahoo Finance	Daily OHLCV stock price data with adjusted close for any ticker	`download_data("stock_prices", symbols = c("AAPL", "MSFT"))`
Index Constituents	Current constituents for 21 major indexes (S&P 500, Russell 2000, DAX, MSCI World, etc.)	`download_data("constituents", index = "DAX")`
Open Source Asset Pricing	213+ monthly asset pricing signals and anomaly variables	`download_data("osap")`
WRDS – CRSP	Monthly and daily stock returns, market cap, exchange, industry, excess returns	`download_data("wrds", "crsp_monthly")`
WRDS – Compustat	Annual and quarterly financial statements with customizable columns	`download_data("wrds", "compustat_annual")`
WRDS – CCM Links	CRSP-Compustat linking table mapping permno to gvkey	`download_data("wrds", "ccm_links")`
WRDS – FISD	Mergent Fixed Income Securities Database with bond characteristics	`download_data("wrds", "fisd")`
WRDS – TRACE	Enhanced TRACE corporate bond transactions with Dick-Nielsen cleaning	`download_data("wrds", "trace_enhanced")`

Table 1: Data sources available via `download_data()` in `tidyfinance` and `py-tidyfinance`.

A programmatic overview of all 318 supported dataset types can be retrieved via `list_supported_types()`, which returns a tibble of domains and datasets. Similarly,

`list_supported_indexes()` returns the names of all supported stock market indexes whose current constituents can be downloaded via the `"constituents"` domain — useful, for instance, to identify the member firms of a benchmark index before constructing portfolios.

2.2. Data cleaning and variable construction

TODO: Describe asset growh: AG year-over-year change in total assets scaled by prior year TODO: Describe major tools and steps for data cleaning and explain how `tidyfinance` helps (particular focus on lagging).

TODO: Explain we remove non-finite values TODO: I removed stock filtering by age (> 2 years) for the sake of simplicity.

```
sorting_variables <- compustat_annual |>
  add_lag_columns(
    "at",
    lag = years(1),
    by = "gvkey"
  ) |>
  rename(at_lag1 = at_lag) |>
  mutate(
    sv_ag = (at - at_lag1) / at_lag1,
    sv_ag = if_else(is.finite(sv_ag), sv_ag, NA_real_),
    filter_earnings = ib
  ) |>
  select(gvkey, date, filter_earnings, sv_ag)
```

TODO: I only show how to do lagging based on FF convention TODO: Make sure to discuss that the same lagging process applies to all sorting variables and that the API allows to retrieve lagged values for all of them.

```
sorting_data <- crsp_monthly |>
  join_lagged_values(
    sorting_variables ,
    "gvkey",
    min_lag = months(12),
    max_lag = months(18),
    ff_adjustment = TRUE
  )
```

2.3. Portfolio sorts

TODO: Explain portfolio sort methodology TODO: Explain `compute_portfolio_returns()` function and its modular components, which allow for flexible portfolio construction based on different methodological choices (e.g., breakpoints, weighting schemes, rebalancing frequency). TODO: Describe the portfolio sort process: defining breakpoints, forming portfolios, and calculating returns. TODO: Discuss the modular components of the portfolio sort process, allowing for customization and flexibility. TODO: Provide examples of different portfolio sort

configurations and their implications for risk factor construction. TODO: Describe in depth the details of sorting: Bunching of sorting variables, aggregation, etc. TODO: Univariate and bivariate sorts. (Bivariate sorts require standardized data output for stock characteristics and can be retrieved from the API). Give a code snippet on how to perform arbitrary wild sorts.

TODO: Tidy finance functions require Bali adjustment and capped value weighting to replace Kelly et al.

Preprocessing choice	Options
Size exclusion	None; bottom 20th NYSE percentile
Exclusion of financials	No; Yes
Exclusion of utilities	No; Yes
Exclusion of neg. earnings	No; Yes
Sorting variable lag	3 months; 6 months; FF
Rebalancing	Monthly; Annual
BP: Quantiles (main)	5; 10
Double sort	Single; Double dependent; Double independent
BP: Quantiles (secondary)	2; 5
BP: Exchanges	NYSE; All
Weighting scheme	EW; VW

Table 2: Preprocessing choices in the portfolio sort grid. For double sorts, the secondary quantiles and exchange choices apply to the size dimension only; for univariate sorts, those rows are not applicable. FF refers to the lag procedure of Fama and French (1993).

TODO: Replace call to grid by proper r-tidyfinance function once implemented TODO: For now I describe the process for one sample subset (exclude_financials is variable), and only for univariate sorts

```
full_grid <- get_available_hf_files("tidy-finance",
      "factor-library-grid"
    )|> pull(url) |>
  arrow::read_parquet()

grid <-  full_grid |>
  filter(sorting_variable == "sv_ag", sorting_variable_lag == "ff",
  exclude_size == 0, exclude_utilities == 0, sorting_method == "univariate")

direction <- "bottom_minus_top"
```

TODO: Explain the necessary columns and what they are needed for

Loop over all remaining tasks and construct factor returns

```
sorting_variable <- "sv_ag"

process_task <- function(row, sorting_data, direction = direction) {
```

```
  if (row$exclude_financials) {
    sorting_data <- sorting_data |>
      filter(industry != "Finance")
  }

  if (row$exclude_negative_earnings) {
    sorting_data <- sorting_data |>
      filter(filter_earnings > 0)
  }

  rebalancing_month <- if (row$rebalancing == "monthly") NULL else 7
  bp_exchanges <- strsplit(row$breakpoints_exchanges, split = "\\|")[[1]]

  options_main <- breakpoint_options(
    n_portfolios = row$breakpoints_main,
    breakpoint_exchanges = bp_exchanges
  )

  min_portfolio_size <- options_main$n_portfolios

  portfolio_returns <- compute_portfolio_returns(
    sorting_data,
    sorting_variable,
    sorting_method = row$sorting_method,
    rebalancing_month,
    options_main,
    min_portfolio_size = min_portfolio_size)

  long_short_return <- portfolio_returns |>
    compute_long_short_returns(direction) |>
    pivot_longer(-date, names_to = "ret_type", values_to = "ret") |>
    mutate(
      id = row$id,
      ret_type = gsub("ret_excess_", "", ret_type),
      sorting_variable = row$sorting_variable,
      ret = replace_na(ret, 0)
    ) |>
    select(id, sorting_variable, date, ret_type, ret)
  return(long_short_return)
}
```

TODO: Explain that we replace portfolio-months with insufficient stocks by zero returns, assuming that the factor would not have been tradable in those month.

TODO: Explain that the factor library contains this output (and all other combinations as well) TBD: The example below shows a socalled specification curve analysis for asset growth. I am not sure it conveys any meaningful economic insights but I lack another appealing example of what to make with results_combined. I think it is important to show some of the results to

make the point that different methodological choices can lead to different factor returns and risk premia.

```
library(patchwork)

specification_data <- results_combined |>
  group_by(id) |>
  summarise(risk_premium = 100 * 12 * mean(ret)) |>
  left_join(full_grid, join_by(id)) |>
  arrange(risk_premium) |>
  mutate(rank = row_number())

decision_cols <- setdiff(names(specification_data), c("id", "sorting_variable", "risk_prem

p_top <- specification_data |>
  ggplot(aes(x = rank, y = risk_premium)) +
  geom_point(size = 0.3, alpha = 0.5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(x = NULL, y = "Annualized risk premium (%)") +
  theme_minimal() +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank())

p_bottom <- specification_data |>
  mutate(across(all_of(decision_cols), as.character)) |>
  pivot_longer(cols = all_of(decision_cols), names_to = "decision", values_to = "choice")
  ggplot(aes(x = rank, y = decision, color = choice)) +
  geom_point(size = 0.3) +
  scale_y_discrete(labels = \(x) gsub("_", " ", x)) +
  labs(x = NULL, y = NULL, color = NULL) +
  theme_minimal() +
  theme(
    axis.text.y = element_text(hjust = 0),
    legend.position = "none"
  )

p_top / p_bottom + plot_layout(heights = c(1, 2))
```

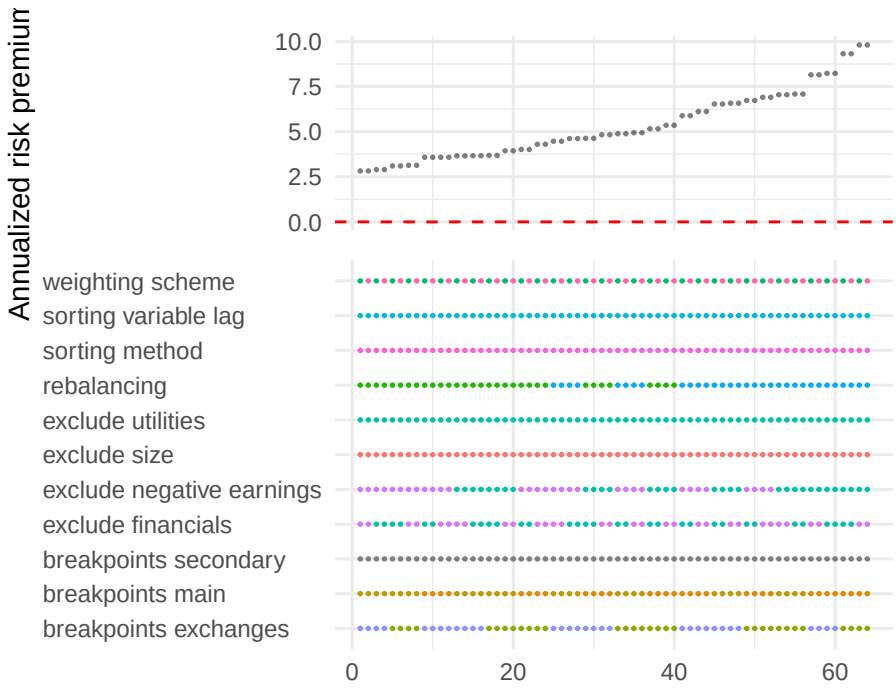


Figure 1: Specification curve for the asset growth factor. Each point in the top panel shows the annualized risk premium for one specification, sorted from lowest to highest. The bottom panel indicates the corresponding modeling choices.

3. Factor library and API

TODO: Describe available sorting variables. TODO: Shiny app for simple download https://app-download-center.cloud.sdu.dk/

Variable	Name		Main reference
52w	52-Week High	↑	George and Hwang (2004)
aci	Abnormal Corporate Investment	↓	Titman, Wei, and Xie (2004)
ami	Amihud Illiquidity	↑	Amihud (2002)
ag	Asset Growth	↓	Cooper, Gulen, and Schill (2008)
beta	Beta	↑	Fama and MacBeth (1973)
bm	Book-to-Market	↑	Daniel and Titman (2006)
cfm	Cash Flow to Market	↑	Lakonishok, Shleifer, and Vishny (1994)
cbop	Cash-Based Operating Profitability	↑	Ball, Gerakos, Linnainmaa, and Nikolaev (2016)
dbe	Change in Common Equity	↓	Richardson, Sloan, Soliman, and Tuna (2005)
dcol	Change in Current Operating Liabilities	↓	Richardson *et al.* (2005)

dfnl	Change in Financial Liabilities	↓	Richardson *et al.* (2005)
dnoa	Change in Net Operating Assets	↓	Hirshleifer, Hou, Teoh, and Zhang (2004)
dpia	Change in PPE and Intangibles	↓	Lyandres, Sun, and Zhang (2008)
dwc	Change in Working Capital	↓	Richardson *et al.* (2005)
cdi	Composite Debt Issuance	↓	Lyandres *et al.* (2008)
csi	Composite Share Issuance	↓	Daniel and Titman (2006)
dm	Debt to Market	↑	Bhandari (1988)
dtv	Dollar Trading Volume	↓	Brennan, Chordia, and Subrahmanyam (1998)
em	Earnings to Market	↑	Basu (1983)
ebm	Enterprise Book to Market	↑	Penman, Richardson, and Tuna (2007)
gpa	Gross Profit Asset	↑	Novy-Marx (2013)
hr	Hiring Rate	↓	Belo and Lin (2012)
iskew	Idiosyncratic Skewness	↓	Bali, Engle, and Murray (2016)
ivol	Idiosyncratic Volatility	↓	Ang, Hodrick, Xing, and Zhang (2006)
dinv	Inventory Changes	↓	Thomas and Zhang (2002)
ig	Investment Growth	↓	Xing (2008)
rev	Long-Term Reversal	↓	De Bondt and Thaler (1985)
me	Market Equity	↓	Banz (1981)
mdr	Maximum Daily Return	↓	Bali, Cakici, and Whitelaw (2011)
mom	Momentum	↑	Fama and French (1996)
ndf	Net Debt Financing	↓	Baker and Wurgler (2006)
ndm	Net Debt to Market	↓	Penman *et al.* (2007)
nef	Net Equity Financing	↓	Baker and Wurgler (2006)
nxf	Net External Financing	↓	Baker and Wurgler (2006)
noa	Net Operating Assets	↓	Hirshleifer *et al.* (2004)
npy	Net Payout Yield	↑	Boudoukh, Michaely, Richardson, and Roberts (2007)
o	O-Score	↓	Ohlson (1980); Dichev (1998)
oa	Operating Accruals	↓	Sloan (1996)
ocm	Operating Cash Flow to Market	↑	Desai, Rajgopal, and Venkatachalam (2004)
ope	Operating Efficiency	↑	Fama and French (2015)
ol	Operating Leverage	↑	Novy-Marx (2011)
rer	Real-Estate Ratio	↑	Titman *et al.* (2004)
rs	Relative Strength	↑	Jegadeesh and Livnat (2006)
rmom	Residual Momentum	↑	Blitz, Huij, and Martens (2011)
roa	Return on Assets	↑	Balakrishnan, Bartov, and Faurel (2010)
roe	Return on Equity	↑	Hou, Xue, and Zhang (2014)
sm	Sales to Market	↑	Barbee Jr, Mukherji, and Raines (1996)
tur	Share Turnover	↓	Datar, Naik, and Radcliffe (1998)
sue	Standardized Unexpected Earnings	↑	Foster, Olsen, and Shevlin (1984)

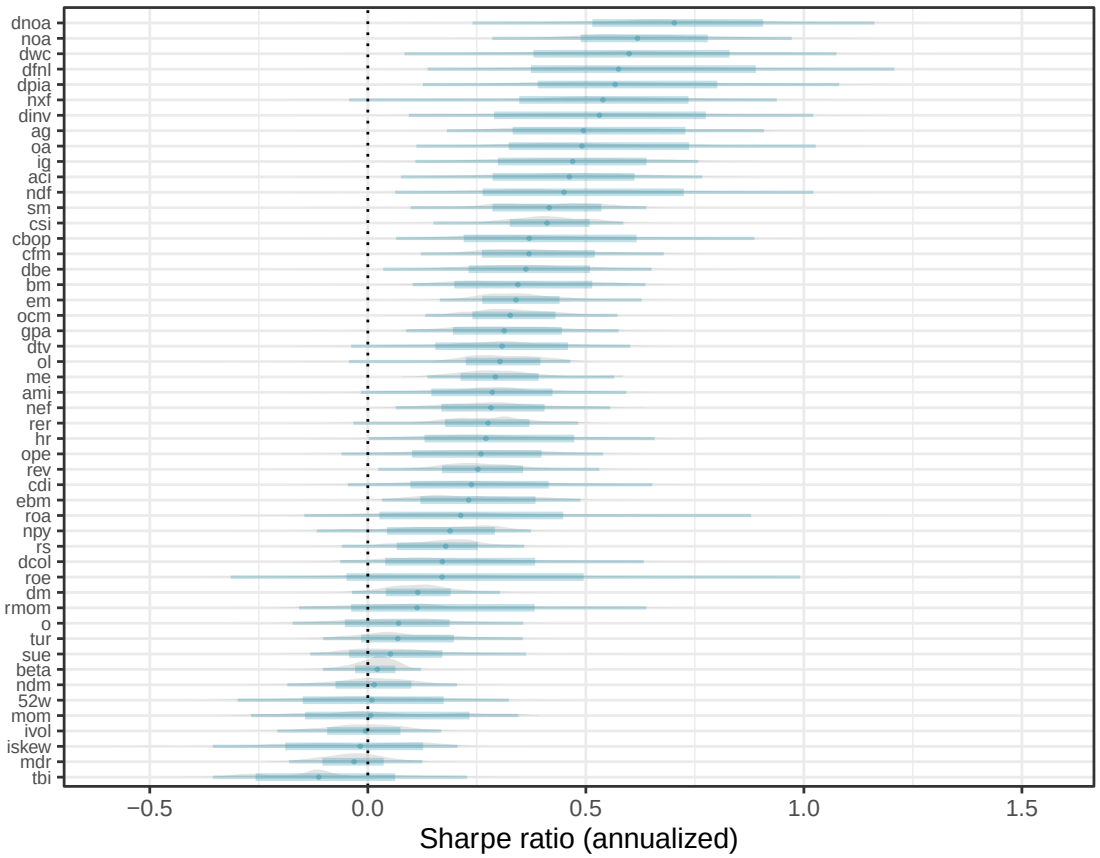


Figure 2: **Sharpe ratios.** The ridge plots indicate the distribution of annualized Sharpe ratios of long-short return differentials for a given sorting variable across all data preprocessing choices illustrated in Table 2. The estimates are based on the available sample period for each factor.

tbi	Taxable Income to Book Income	↓	Lev and Nissim (2004)

Table 3: Sorting variables included in the factor library. The direction column indicates whether the long-short factor return is constructed as top-minus-bottom (↑) or bottom-minus-top (↓) of the characteristic distribution.

```
Warning: There was 1 warning in `mutate()`.
i In argument: `sorting_variable = forcats::fct_reorder(...)`.
Caused by warning:
! `fct_reorder()` removing 9980 missing values.
i Use `.na_rm = TRUE` to silence this message.
i Use `.na_rm = FALSE` to preserve NAs.
```

```
Warning: Removed 9980 rows containing missing values or values outside the scale range
(`stat_slabinterval()`).
```

4. Summary and discussion

Computational details

TODO: Fill out details on the computational environment, including software versions, packages used, and any relevant hardware specifications. This section should provide enough information for others to replicate the computational aspects of the research. TODO: Reference to github repository with all code and data workflows.

- The results in this paper were obtained using R~4.5.2 with the **MASS**~7.3.47 package. R itself and all packages used are available from the Comprehensive R Archive Network (CRAN) at
- [https://CRAN.R-project.org/]. Python environment details.

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References

Amihud Y (2002). "Illiquidity and stock returns: cross-section and time-series effects." *Journal of Financial Markets*, **5**(1), 31–56.

Ang A, Hodrick RJ, Xing Y, Zhang X (2006). "The cross-section of volatility and expected returns." *The Journal of Finance*, **61**(1), 259–299.

Baker M, Wurgler J (2006). "Investor sentiment and the cross-section of stock returns." *The Journal of Finance*, **61**(4), 1645–1680.

Balakrishnan K, Bartov E, Faurel L (2010). "Post loss/profit announcement drift." *Journal of Accounting and Economics*, **50**(1), 20–41.

Bali TG, Cakici N, Whitelaw RF (2011). "Maxing out: Stocks as lotteries and the cross-section of expected returns." *Journal of Financial Economics*, **99**(2), 427–446.

Bali TG, Engle RF, Murray S (2016). *Empirical asset pricing: The cross section of stock returns*. John Wiley & Sons.

Ball R, Gerakos J, Linnainmaa JT, Nikolaev V (2016). "Accruals, cash flows, and operating profitability in the cross section of stock returns." *Journal of Financial Economics*, **121**(1), 28–45.

Banz RW (1981). "The relationship between return and market value of common stocks." *Journal of Financial Economics*, **9**(1), 3–18.

Barbee Jr WC, Mukherji S, Raines GA (1996). "Do sales–price and debt–equity explain stock returns better than book–market and firm size?" *Financial Analysts Journal*, **52**(2), 56–60.

Basu S (1983). "The relationship between earnings' yield, market value and return for NYSE common stocks: Further evidence." *Journal of Financial Economics*, **12**(1), 129–156.

Belo F, Lin X (2012). "The inventory growth spread." *The Review of Financial Studies*, **25**(1), 278–313.

Bhandari LC (1988). "Debt/equity ratio and expected common stock returns: Empirical evidence." *The Journal of Finance*, **43**(2), 507–528.

Blitz D, Huij J, Martens M (2011). "Residual momentum." *Journal of Empirical Finance*, **18**(3), 506–521.

Boudoukh J, Michaely R, Richardson M, Roberts MR (2007). "On the importance of measuring payout yield: Implications for empirical asset pricing." *The Journal of Finance*, **62**(2), 877–915.

Brennan MJ, Chordia T, Subrahmanyam A (1998). "Alternative factor specifications, security characteristics, and the cross-section of expected stock returns." *Journal of Financial Economics*, **49**(3), 345–373.

Chen A, Zimmermann T (2022). "Open Source Cross-Sectional Asset Pricing." *Critical Finance Review*, **27**(2), 207–264.

Cochrane JH (2011). "Presidential address: Discount rates." *The Journal of Finance*, **66**(4), 1047–1108.

Cooper MJ, Gulen H, Schill MJ (2008). "Asset growth and the cross-section of stock returns." *The Journal of Finance*, **63**(4), 1609–1651.

Daniel K, Titman S (2006). "Market reactions to tangible and intangible information." *The Journal of Finance*, **61**(4), 1605–1643.

Datar VT, Naik NY, Radcliffe R (1998). "Liquidity and stock returns: An alternative test." *Journal of Financial Markets*, **1**(2), 203–219.

Davis JL, Fama EF, French KR (2000). "Characteristics, covariances, and average returns: 1929 to 1997." *The Journal of Finance*, **55**(1), 389–406.

De Bondt WF, Thaler R (1985). "Does the stock market overreact?" *The Journal of Finance*, **40**(3), 793–805.

Desai H, Rajgopal S, Venkatachalam M (2004). "Value-glamour and accruals mispricing: One anomaly or two?" *The Accounting Review*, **79**(2), 355–385.

Dichev ID (1998). "Is the risk of bankruptcy a systematic risk?" *The Journal of Finance*, **53**(3), 1131–1147.

Fama EF, French KR (1992). "The cross-section of expected stock returns." *The Journal of Finance*, **47**(2), 427–465.

Fama EF, French KR (1993). "Common risk factors in the returns on stocks and bonds." *Journal of Financial Economics*, **33**(1), 3–56.

Fama EF, French KR (1996). "Multifactor explanations of asset pricing anomalies." *The Journal of Finance*, **51**(1), 55–84.

Fama EF, French KR (2015). "A five-factor asset pricing model." *Journal of Financial Economics*, **116**(1), 1–22.

Fama EF, French KR (2023). "Production of U.S. Rm-Rf, SMB, and HML in the Fama-French Data Library." Chicago Booth Research Paper No. 23-22, Fama-Miller Working Paper. Available at SSRN: https://ssrn.com/abstract=4629613 or DOI: http://dx.doi.org/10.2139/ssrn.4629613.

Fama EF, MacBeth JD (1973). "Risk, return, and equilibrium: Empirical tests." *Journal of Political Economy*, **81**(3), 607–636.

Foster G, Olsen C, Shevlin T (1984). "Earnings releases, anomalies, and the behavior of security returns." *The Accounting Review*, **59**(4), 574–603.

Gao X, Ritter JR (2010). "The marketing of seasoned equity offerings." *Journal of Financial Economics*, **97**(1), 33–52.

George TJ, Hwang CY (2004). "The 52-week high and momentum investing." *The Journal of Finance*, **59**(5), 2145–2176.

Green J, Hand JR, Zhang XF (2013). "The supraview of return predictive signals." *Review of Accounting Studies*, **18**(3), 692–730.

Harvey CR, Liu Y, Zhu H (2016). "… and the Cross-Section of Expected Returns." *The Review of Financial Studies*, **29**(1), 5–68. doi:10.1093/rfs/hhv059.

Hirshleifer D, Hou K, Teoh SH, Zhang Y (2004). "Do investors overvalue firms with bloated balance sheets?" *Journal of Accounting and Economics*, **38**, 297–331.

Hou K, Xue C, Zhang L (2014). "Digesting Anomalies: An Investment Approach." *The Review of Financial Studies*, **28**(3), 650–705.

Hou K, Xue C, Zhang L (2020). "Replicating anomalies." *The Review of Financial Studies*, **33**(5), 2019–2133.

Hribar P, Collins DW (2002). "Errors in estimating accruals: Implications for empirical research." *Journal of Accounting Research*, **40**(1), 105–134.

Jegadeesh N, Livnat J (2006). "Revenue surprises and stock returns." *Journal of Accounting and Economics*, **41**(1-2), 147–171.

Jensen TI, Kelly B, Pedersen LH (2023). "Is there a replication crisis in finance?" *The Journal of Finance*, **78**(5), 2465–2518.

Lakonishok J, Shleifer A, Vishny RW (1994). "Contrarian investment, extrapolation, and risk." *The Journal of Finance*, **49**(5), 1541–1578.

Lev B, Nissim D (2004). "Taxable income, future earnings, and equity values." *The Accounting Review*, **79**(4), 1039–1074.

Lyandres E, Sun L, Zhang L (2008). "The new issues puzzle: Testing the investment-based explanation." *The Review of Financial Studies*, **21**(6), 2825–2855.

Novy-Marx R (2011). "Operating leverage." *Review of Finance*, **15**(1), 103–134.

Novy-Marx R (2013). "The other side of value: The gross profitability premium." *Journal of Financial Economics*, **108**(1), 1–28.

Ohlson JA (1980). "Financial ratios and the probabilistic prediction of bankruptcy." *Journal of Accounting Research*, **18**(1), 109–131.

Penman SH, Richardson SA, Tuna I (2007). "The book-to-price effect in stock returns: accounting for leverage." *Journal of Accounting Research*, **45**(2), 427–467.

Richardson SA, Sloan RG, Soliman MT, Tuna I (2005). "Accrual reliability, earnings persistence and stock prices." *Journal of Accounting and Economics*, **39**(3), 437–485.

Sloan RG (1996). "Do stock prices fully reflect information in accruals and cash flows about future earnings?" *The Accounting Review*, **71**(3), 289–315.

Thomas JK, Zhang H (2002). "Inventory changes and future returns." *Review of Accounting Studies*, **7**(2), 163–187.

Titman S, Wei KJ, Xie F (2004). "Capital investments and stock returns." *Journal of Financial and Quantitative Analysis*, **39**(4), 677–700.

Xing Y (2008). "Interpreting the value effect through the Q-theory: An empirical investigation." *The Review of Financial Studies*, **21**(4), 1767–1795.

Technical details

Internet Appendix

Variable construction

In this section, we show the construction details for the risk factors. We group them by their underlying mechanism as shown in Table 3 in the Appendix of the main paper.

IA-1. Financing

CDI - Composite Debt Issuance

Following Lyandres *et al.* (2008), we compute composite debt issuance as the logarithmic growth rate of book debt (i.e., current debt, DLC, plus long-term debt, DLTT) over the last five years, i.e.,

$$\text{CDI}_t = \log(\text{DLC}_t + \text{DLTT}_t) - \log(\text{DLC}_{t-60} + \text{DLTT}_{t-60})$$

We replace the missing values of DLC and DLTT with zero individually. Additionally, the sum of DLC and DLTT must be larger than zero, and is set to missing otherwise.

CSI - Composite Share Issuance

Following Daniel and Titman (2006), we measure composite share issuance from CRSP data as the difference between the change in market equity (ME) (i.e., the shares outstanding, shrout, times the absolute value of their price, altprc, in million USD) and the cumulative log return (where r is a simple return) in each month from t to $t-60$, i.e.,

$$\text{CSI}_t = \log(ME_t/ME_{t-60}) - \sum_{\tau=t-59}^{t} \log(1+r_\tau)$$

DBE - Change in Common Equity

Following Richardson *et al.* (2005), we calculate the change in common equity as the change in common equity (CEQ) over one year scaled by lagged assets (AT), i.e.,

$$DBE_t = \frac{CEQ_t - CEQ_{t-12}}{AT_{t-12}}$$

DCOL - Change in Current Operating Liabilities

Following Richardson *et al.* (2005), we measure the change in current operating liabilities as the change in current operating liabilities (i.e., current liabilities, LCT, minus current debt, DLC) over one year scaled by lagged assets (AT), i.e.,

$$DCOL_t = \frac{(LCT_t - DLC_t) - (LCT_{t-12} - DLC_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC with 0.

DFNL - Change in Financial Liabilities

Following Richardson *et al.* (2005), we measure the change in financial liabilities (i.e., the sum of current debt, DLC, plus long-term debt, DLTT, plus preferred stocks, PSTK) over one year scaled by lagged assets (AT), i.e.,

$$DFNL_t = \frac{(DLC_t + DLTT_t + PSTK_t) - (DLC_{t-12} + DLTT_{t-12} + PSTK_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC, DLTT and PSTK with 0, if at least one of them is given.

NDF - Net Debt Financing

Following Baker and Wurgler (2006), we measure net debt financing as cash proceeds from the issuance of long-term debt (DLTIS) minus cash payments for long-term debt reductions (DLTR) plus net changes in current debt (DLCCH) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NDF_t = \frac{DLTIS_t - DLTR_t + DLCCH_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

We replace the missing values of DLCCH with zero.

NEF - Net Equity Financing

Following Baker and Wurgler (2006), we measure net equity financing as proceeds from the sale of common and preferred stocks (SSTK) minus payments for the repurchase of common and preferred stocks (PRSTKC) minus cash payments for dividends (DV) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NEF_t = \frac{SSTK_t - PRSTKC_t - DV_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

NXF - Net External Financing

Following Baker and Wurgler (2006), we measure net external financing as the sum of net debt financing (NDF) and net equity financing (NEF) as defined above.

IA-2. Intangibles

HR - Hiring Rate

Following Belo and Lin (2012), we measure the hiring rate as the change in the number of employees (EMP) over one year scaled by the average number of employees, i.e.,

$$HR_t = \frac{EMP_t - EMP_{t-12}}{\frac{1}{2}(EMP_t + EMP_{t-12})}$$

We exclude firms with a hiring rate of zero.

OL - Operating Leverage

Following Novy-Marx (2011), we measure operating leverage as cost of goods sold (COGS) plus selling, general, and administrative expenses (XSGA), both scaled by current total assets (AT), i.e.,

$$OL_t = \frac{COGS_t + XSGA_t}{AT_t}$$

RER - Real-Estate Ratio

Following Titman *et al.* (2004), we measure the real-estate ratio in two subperiods differently.

(a) Prior to 1983, the real-estate ratio is the sum of buildings (PPENB) and capital leases (PPENLS) scaled by net property, plant, and equipment (PPENT), i.e.,

$$RER_t = \frac{PPENB_t + PPENLS_t}{PPENT_t}$$

(b) After the end of 1983, the real-estate ratio is the sum of buildings at cost (FATB) and leases at cost (FATL), both divided by gross property, plant, and equipment (PPEGT), i.e.,

$$RER_t = \frac{FATB_t + FATL_t}{PPEGT_t}$$

Subsequently, we winsorize the real-estate ratios in each fiscal year at the first and 99th percentile. The industry-adjusted real-estate ratio is obtained by subtracting the industry average real-estate ratio from each stock-specific real-estate ratio. We use 2-digit SIC codes to assign stocks to industries. We always require at least five observations to calculate the industry average each year.

IA-3. Investment

ACI - Abnormal Corporate Investment

Following Titman *et al.* (2004), we measure abnormal corporate investments using the variable CE, which is capital expenditures (CAPX) scaled by sales (SALE), then ACI is given by CE scaled by CE from the previous three years minus one, i.e.,

$$ACI_t = \frac{CE_t}{\frac{1}{3}(CE_{t-12} + CE_{t-24} + CE_{t-36})} - 1$$

We follow Hou, Xue, and Zhang (2020) and exclude stocks with sales below 10 million dollars.

AG - Asset Growth

Following Cooper *et al.* (2008), we measure asset growth as the change in total assets (AT) over one year scaled by lagged assets (AT), i.e.,

$$AG_t = \frac{AT_t - AT_{t-12}}{AT_{t-12}}$$

DINV - Inventory Changes

Following Thomas and Zhang (2002), we measure the change in inventory as the annual change in inventories (INVT) scaled by average total assets (AT) over the current and the last year, i.e.,

$$DINV_t = \frac{INVT_t - INVT_{t-12}}{\frac{1}{2}(AT_t + AT_{t-12})}$$

DNOA - Change in Net Operating Assets

Following Hirshleifer *et al.* (2004), we measure net operating assets (defined as NOA^{int}) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, the change in net operating assets is the difference between net operating assets of fiscal year t and fiscal year $t-12$ scaled by lagged assets (AT), i.e.,

$$DNOA_t = \frac{NOA_t^{int} - NOA_{t-12}^{int}}{AT_{t-12}}$$

DPIA - Change in Property, Plant, Equipment, and Inventory

Following Lyandres *et al.* (2008), we measure the change in property, plant, equipment, and inventory by combining the change in gross property, plant, and equipment (PPEGT) with the annual change in inventory (INVT) and scaling this sum by lagged total assets (AT), i.e.,

$$DPIA_t = \frac{(PPEGT_t - PPEGT_{t-12}) + (INVT_t - INVT_{t-12})}{AT_{t-12}}$$

DWC - Change in Net Non-Cash Working Capital

Following Richardson *et al.* (2005), we measure non-cash working capital as

$$WC_t = (ACT_t - CHE_t) - (LCT_t - DLC_t)$$

where ACT corresponds to current assets, CHE to cash, LCT to current liabilities, and DLC to short-term debt. We set missing values of DLC to zero. Then, the change in net non-cash

working capital corresponds to the change in WC from fiscal year $t-12$ to fiscal year t scaled by lagged total assets (AT), i.e.,

$$DWC_t = \frac{WC_t - WC_{t-12}}{AT_{t-12}}$$

IG - Investment Growth

Following Xing (2008), we measure investment growth as the annual change in capital expenditures (CAPX) scaled by lagged capital expenditures, i.e.,

$$IG_t = \frac{CAPX_t - CAPX_{t-12}}{CAPX_{t-12}}$$

NOA - Net Operating Assets

Following Hirshleifer *et al.* (2004), we measure net operating assets (defined as NOA^{int}) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, net operating assets is the net operating assets of fiscal year t scaled by lagged assets (AT), i.e.,

$$NOA_t = \frac{NOA_t^{int}}{AT_{t-12}}$$

OA - Operating Accruals

Following Sloan (1996) (before 1988) and Hribar and Collins (2002) (after 1988), we measure operating accruals until 1988 as

$$OA_t = \frac{(\Delta ACT_t - \Delta CHE_t) - (\Delta LCT_t - \Delta DLC_t - \Delta TXP_t) - DP_t}{AT_{t-12}}$$

where ACT is current assets, CHE cash, LCT current liabilities, DLC short-term debt, TXP taxes payable, and DP depreciation and amortization. Moreover, we replace the missing values of DLC and TXP with zero. Δ represents changes over one year. From 1988 onwards, we measure operating accruals as

$$OA_t = \frac{NI_t - OANCF_t}{AT_{t-12}}$$

where NI is net income and OANCF corresponds to net cash flow from operations.

IA-4. Momentum

MOM - Return Momentum

Following Fama and French (1996), we measure return momentum from monthly CRSP returns (where r is a simple return) as the cumulative return from month $t-11$ to month $t-1$ skipping the most recent month, i.e.,

$$\text{MOM}_t = \exp\left(\sum_{\tau=t-11}^{t-1} \log(1+r_\tau)\right) - 1$$

RMOM - Residual Momentum

Following Blitz *et al.* (2011), we define the 11-month residual momentum (RMOM) in each month and for each stock as cumulative residual returns from month $t-11$ to month $t-1$, scaled by the standard deviation of residual returns over the same time horizon. Residual returns ϵ are obtained in each month from regressing monthly simple, excess stock returns from month $t-1$ to month $t-36$ on the Fama and French (1993) three-factor model. Throughout these rolling regressions, we always require 36 monthly returns.

RS - Revenue Surprise

Following Jegadeesh and Livnat (2006), we measure the change in revenues per share (DRPS) from quarterly Compustat data as quarterly sales (SALEQ) scaled by the product of quarterly shares outstanding (CSHPRQ) times their cumulative adjustment factor (AJEXQ), i.e.,

$$DRPS_q = \frac{SALEQ_q}{CSHPRQ_q \cdot AJEXQ_q} - \frac{SALEQ_{q-4}}{CSHPRQ_{q-4} \cdot AJEXQ_{q-4}}$$

Then, the revenue surprise corresponds to the change in revenues per share over the last four quarters scaled by the standard deviation of the change in revenues per share over the last eight quarters. We require at least six quarterly observations for this rolling standard deviation and remove infinite values. Finally, the earnings announcement date has to be after the fiscal quarter end date.

SUE - Standardized Unexpected Earnings

Following Foster *et al.* (1984), we measure unexpected earnings per share (UES) from quarterly Compustat data as the change in split-adjusted earnings (which are equal to quarterly earnings per share, EPSPXQ, scaled by the correction factor for shares outstanding, AJEXQ) per share from its value four quarters ago

$$UES_q = \frac{EPSPXQ_q}{AJEXQ_q} - \frac{EPSPXQ_{q-4}}{AJEXQ_{q-4}}$$

Then, standardized unexpected earnings are defined as unexpected earnings per share divided by the standard deviation of unexpected earnings per share over the previous eight quarters. We require at least six quarterly observations for this rolling standard deviation. Finally, the earnings announcement date has to be after the fiscal quarter end date.

52W - 52-Week High

Following George and Hwang (2004), we measure the 52-week high as month t's closing split-adjusted stock price scaled by the highest daily split-adjusted stock price over the previous twelve months. We only consider stock-month pairs with more than three valid price observations.

REV - Long-Term Reversal

Following De Bondt and Thaler (1985), we measure the long-term reversal effect by the cumulative returns from month $t-60$ to month $t-13$, i.e.,

$$\text{REV}_t = \exp\left(\sum_{\tau=t-60}^{t-13} \log(1+r_\tau)\right) - 1$$

IA-5. Profitability

CBOP - Cash-Based Operating Profitability

Following Ball *et al.* (2016), we measure cash-based operating profitability

$$\begin{aligned} CBOP_t =& (REVT_t - COGS_t - XSGA_t + XRD_t \\ & - \Delta RECT_t - \Delta INVT_t - \Delta XPP_t + \Delta DRC_t \\ & + \Delta DRLT_t + \Delta AP_t + \Delta XACC_t)/AT_t \end{aligned}$$

where REVT is total revenue, COGS are cost of goods sold, XSGA are selling, general, and administrative expenses, and XRD are R&D expenses. Moreover, Δ refers to one-year changes, which we measure for accounts receivable (RECT), inventory (INVT), prepaid expenses (XPP), deferred revenues (the sum of current, DRC, and long-term, DRLT), trade accounts payable (AP), and accrued expenses (XACC). We set the missing values of XRD and all missing changes to zero.

GPA - Gross Profits to Assets

Following Novy-Marx (2013), we measure gross profits to assets as total revenues (REVT) minus cost of goods sold (COGS) scaled by current total assets (AT), i.e.,

$$GPA_t = \frac{REVT_t - COGS_t}{AT_t}$$

O - Ohlson's O-score

Following Ohlson (1980), we measure O-score with the following linear relation

$$
\begin{aligned}
O_t = &- 1.32 - 0.407 \cdot \log(AT_t) + 6.03 \cdot \frac{DLC_t + DLTT_t}{AT_t} \\
&- 1.43 \cdot \frac{ACT_t - LCT_t}{AT_t} + 0.076 \cdot \frac{LCT_t}{AT_t} \\
&- 1.72 \cdot \mathbf{1}_{LT_t > AT_t} - 2.37 \cdot \frac{NI_t}{AT_t} - 1.83 \cdot \frac{PI_t + DP_t}{LT_t} \\
&+ 0.285 \cdot \mathbf{1}_{NI_t<0 \text{ and } NI_{t-12}<0} - 0.521 \cdot \frac{NI_t - NI_{t-12}}{|NI_t| + |NI_{t-12}|}
\end{aligned}
$$

where AT corresponds to total assets, DLC to short-term debt, DLTT to long-term debt, ACT to current assets, LCT to current liabilities, LT to total liabilities, PI to pretax income, DP to depreciation and amortization, and NI to net income. We winsorize all variables except for dummy variables at the first and 99th percentile of their respective distribution.

OPE - Operating Profits to Book Equity

Following Fama and French (2015), we measure operating profits to book equity as

$$
OPE_t = \frac{REVT_t - COGS_t - XSGA_t - XINT_t}{BE_t}
$$

where REVT corresponds to total revenues, COGS to cost of goods sold, XSGA to selling, general and administrative expenses, XINT to interest expenses, and BE to book equity. The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero. Moreover, missing values in COGS, XSGA, and XINT are set to zero.

ROA - Return on Assets

Following Balakrishnan *et al.* (2010), we measure return on assets as quarterly income before extraordinary items (IBQ) relative to lagged quarterly total assets (ATQ), i.e.,

$$
ROA_q = \frac{IBQ_q}{ATQ_{q-1}}
$$

Finally, the earnings announcement date has to be after the fiscal quarter end date.

ROE - Return on Equity

Following Hou *et al.* (2014), we measure return on equity as quarterly income before extraordinary items (IBQ) relative to lagged book equity (BEQ)

$$ROE_q = \frac{IBQ_q}{BEQ_{q-1}}$$

Quarterly book equity (BEQ) is computed as the (a) book equity of shareholders plus (b) balance sheet deferred taxes and investment tax credit minus (c) the book value of preferred stock. Regarding (a), we measure book equity of shareholders by shareholders' equity (SEQQ), or the sum of common equity (CEQQ) and the par value of preferred stock (PSTKQ), or if all previous items are unavailable by total assets (ATQ) minus total liabilities (LTQ). Regarding (b), the balance sheet deferred taxes and investment tax credit is TXDITCQ, or TXDBQ, or zero if both are missing. Regarding (c), the book value of preferred stocks corresponds to PSTKRQ, or PSTKQ, or zero if both are unavailable. Finally, the earnings announcement date has to be after the fiscal quarter end date.

TBI - Taxable Income to Book Income

Following Green, Hand, and Zhang (2013), we measure taxable income to book income as pretax income (PI) scaled by net income (NI)

$$TBI_t = \frac{PI_t}{NI_t}$$

We require positive pretax and net income.

IA-6. Size

ME - Size

Following Fama and French (1992), we measure size as the natural logarithm of the market equity. We obtain market equity data from CRSP by multiplying the shares outstanding (shrout * 1,000, as they are given in thousands) with the corresponding share price (the absolute value of altprc), i.e.,

$$ME_t = \log\left(shrout_t * 1000 * |altprc_t|\right)$$

IA-7. Trading frictions

AMI - Amihud Illiquidity Measure

Following Amihud (2002), we measure the illiquidity from daily return-to-volume (RV), which is the absolute daily return (r) scaled by the daily dollar trading volume (product of vol and prc), i.e.,

$$RV_t = \frac{|r_d|}{prc_t * vol_t}$$

Then, the Amihud illiquidity measure corresponds to the average return to volume estimate over the last six months. We require at least 50 observations for this average and adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

BETA - Beta Relative to the Market

Following Fama and MacBeth (1973), we measure the market beta β_1 from monthly CRSP data. Specifically, we run the following time series regression over the previous five years

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + u_t$$

Moreover, we require at least 24 monthly observations for the regression above.

DTV - Dollar Trading Volume

Following Brennan *et al.* (1998), we measure dollar trading volume from daily CRSP data as the average dollar trading volume (product of vol and prc) from month $t-6$ to month $t-1$ (with at least 50 observations). Moreover, we adjust dollar trading volume from NASDAQ according to Gao and Ritter (2010).

ISKEW - Idiosyncratic Skewness Relative to the Fama and French (1993) three-factor model

Following Bali *et al.* (2016), we regress daily excess returns in each month on the three Fama and French (1993) factor model (with a minimum of 15 daily observations), i.e.,

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

Then, idiosyncratic skewness is measured as the skewness of residuals u_t.

IVOL - Idiosyncratic Volatility Relative to the Fama and French (1993) model

Following Ang *et al.* (2006), we compute idiosyncratic volatility relative to the Fama and French (1993) three-factor model as the volatility of residuals from the following regression (with a minimum of 15 daily observations)

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

MDR - Maximum Daily Return

Following Bali *et al.* (2011), we measure the maximum daily (excess) return in each month from daily CRSP data (with at least 15 return observations).

TUR - Share Turnover

Following Datar *et al.* (1998), we measure the daily share turnover of each stock as the number of shares traded (vol) scaled by the number of shares outstanding (shrout), i.e.,

$$T_t = \frac{vol}{shrout}$$

Then, the share turnover for each firm in each month t is the average daily share turnover over the previous six months (with at least 50 observations). Moreover, we adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

IA-8. Valuation

BM - Book Equity to Market Equity

Following Davis, Fama, and French (2000), we measure the book-to-market ratio as book equity (BE) from Compustat divided by market equity (ME) from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$BM_t = \frac{BE_t}{ME_t}$$

The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero.

CFM - Cash Flow to Market Equity

Following Lakonishok *et al.* (1994), we measure the cash-flow-to-price ratio as income before extraordinary items (IB) plus depreciation (DP) scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$CFM_t = \frac{IB_t + DP_t}{ME_t}$$

We exclude all stocks with negative cash flows.

DM - Debt to Market Equity

Following Bhandari (1988), we measure the debt-to-market ratio as short-term debt (DLC) plus long-term debt (DLTT) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$DM_t = \frac{DLC_t + DLTT_t}{ME_t}$$

We exclude stocks with missing DLC and DLTT observations.

EBM - Enterprise Book Equity to Market Equity

Following Penman *et al.* (2007), we measure enterprise book equity scaled by market equity as net debt plus book equity scaled by net debt plus market equity

$$EBM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + BE_t}{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments, and ME to the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD). Lastly, in this case, book equity BE is computed as common equity (CEQ) plus TSTKP minus DVPA. Missing observations in DVPA and TSTKP are set to zero. Additionally, we require that the sum of net debt and book equity, as well as the sum of net debt plus market equity, are positive.

EM - Earnings to Market Equity

Following Basu (1983), we measure the earnings-to-price ratio as income before extraordinary items (IB) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$EM_t = \frac{IB_t}{ME_t}$$

We exclude firms with negative earnings.

NDM - Net Debt to Market Equity

Following Penman *et al.* (2007), we measure net debt to price as net debt from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$NDM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t}{ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments. Lastly, missing observations in DVPA and TSTKP are set to zero.

NPY - Net Payout Yield

Following Boudoukh *et al.* (2007), we measure the net payout yield as net payouts from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$\begin{aligned} NPY_t = &[(DVC_t + PRSTKC_t + \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV<0}) \\ &- (SSTK_t - \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV>0})]/ME_t \end{aligned}$$

where DVC are dividends from common stock, PRSTKC is the purchase of common and preferred stock, PSTKRV is the value of the net number of preferred stocks outstanding, and SSTK reflects the sale of common and preferred stocks. $1_{\Delta PSTKRV<0}$ is a dummy variable that has value one if the annual change in PSTKRV is negative and zero otherwise. Moreover, we exclude stocks with negative net payouts.

OCM - Operating Cash Flow to Market Equity

Following Desai *et al.* (2004), we measure the ratio of operating cash flows to price as operating cash flows (OC) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$OCM_t = \frac{OC_t}{ME_t}$$

Before 1988, we measure OC as funds from operations (FOPT) minus the change in working capital (item WCAP). Thereafter, we measure OC as net cash flows from operating activities (OANCF). Moreover, we exclude firms with negative operating cash flows.

SM - Sales to Market Equity

Following Barbee Jr *et al.* (1996), we measure the sales-to-price ratio (SM) as sales (SALE) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$SM_t = \frac{SALE_t}{ME_t}$$

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